

10. Drone

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10.1. Introduction

10.1.1. Project Overview

This comprehensive documentation details the development and implementation of an autonomous drone navigation system for the MORPHEUS Rover Project, incorporating feedback from mid-term and final presentations. The project focuses on leveraging behavioral cloning and reinforcement learning techniques through MATLAB to enhance drone capabilities for the European Rover Challenge (ERC). The system enables drones to learn from expert demonstrations and replicate complex navigation patterns, significantly improving operational effectiveness in GPS-denied environments while maintaining seamless coordination with the rover team.

The MORPHEUS Rover Project integrates advanced navigation techniques, sensor fusion, and machine learning to create a robust drone system capable of autonomous operation in challenging environments. The project emerged from a gap analysis that identified limited adaptability in novel environments as a key shortcoming in current drone navigation systems.

The drone team's approach incorporates:

- GPS-denied navigation using LiDAR and Visual Inertial Odometry (VIO)
- Real-time obstacle detection using YOLO (You Only Look Once)
- Continuous communication via 4G LTE ground station
- Behavioral cloning to replicate expert pilot behavior
- Integration with rover operations for collaborative exploration
- Extensive MATLAB implementation for simulation, data processing, and control algorithms

The system is specifically designed to handle the "worst case" scenario of detecting green probes on green grass during competition tasks, requiring sophisticated image processing and machine learning techniques to overcome this challenging visual discrimination problem.

10.2. MATLAB Integration

10.2.1. MATLAB Usage Requirements

Following feedback from presentations, extensive MATLAB implementation is a central requirement for this project. MATLAB serves as the primary development environment for multiple system components. The project mandates maximizing MATLAB usage across all components, as emphasized in the feedback:

- Implementation score is based on the extent of MATLAB utilization in the final report.
- While other tools are permitted, MATLAB should be prioritized whenever technically feasible.
- Training, tutorials, licenses, and white papers will be provided to support extensive MATLAB integration.

10.2.2. MATLAB Implementation Areas

Key areas of MATLAB implementation include:

- **Drone Dynamics Modeling:** Developing 6DOF (six degrees of freedom) models for accurate flight simulation.
- **Control System Design:** Implementing PID controllers and advanced control algorithms.
- **Sensor Data Processing:** Processing LiDAR, camera, and IMU data.
- **Machine Learning Pipeline:** Training and deploying behavioral cloning and reinforcement learning models.
- **Path Planning:** Generating optimal trajectories for navigation and probe detection.
- **Ground Station Interface:** Creating visualization tools and data displays.
- **Post-Processing:** Analyzing flight data and mission performance.

MATLAB's UAV Toolbox provides specific capabilities for drone development, including modeling UAV architecture, designing flight controllers, developing perception systems, planning motion, and generating production code for deployment. These capabilities are leveraged extensively throughout the project.

10.3. System Architecture

10.3.1. Record-and-Replay Architecture

The core behavioral cloning system uses a record-and-replay architecture implemented in MATLAB that enables the drone to reproduce movements originally performed by a human pilot:

- **RC Controller to Receiver:** A handheld joystick/RC transmitter sends control commands wirelessly to a FlySky FS A8S 8-channel receiver.
- **Receiver to Arduino:** The FS A8S receiver outputs an 8-channel PPM signal, read by an Arduino microcontroller.
- **Arduino and PC Recording/Replaying:** The Arduino streams the control data to a PC running MATLAB for storage during recording. During replay, the Arduino requests the saved data from the PC.
- **Arduino to Crossfire Transmitter:** The Arduino forwards control signals to a TBS Crossfire transmitter module.
- **Crossfire to Drone:** The Crossfire transmitter sends the control signals to the drone's onboard Crossfire receiver.

MATLAB handles the computational aspects including data logging, preprocessing, model training, and control signal generation. This approach allows for expert pilot behaviors to be accurately captured, processed, and reproduced.

10.3.2. Behavioral Cloning Schematic

The behavioral cloning system follows a clear signal path from input to output:

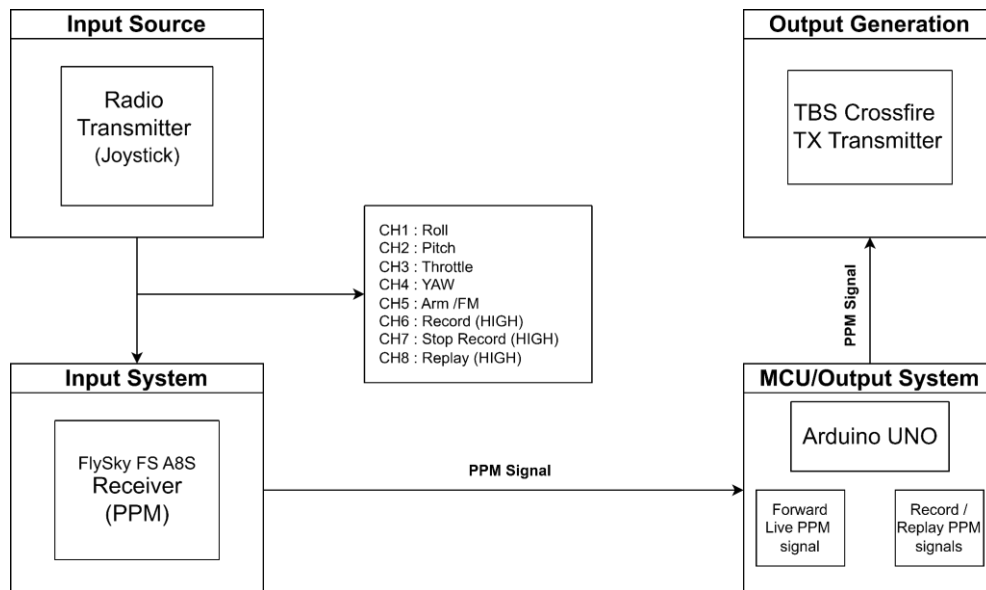


Figure 99: Behavioral Cloning System Chart

- **Input Source:** Radio transmitter (joystick) provides control inputs.
- **Input System:** FlySky FS A8S receiver processes PPM signals.
- **MCU/Output System:** Arduino UNO serves as the central processing unit.
- **Output Generation:** TBS Crossfire TX transmitter sends commands to the drone.

The system manages eight channels:

- CH1: Roll
- CH2: Pitch
- CH3: Throttle
- CH4: YAW
- CH5: Arm/Flight Mode
- CH6: Record HIGH
- CH7: Stop Record HIGH
- CH8: Replay/Mission Trigger HIGH

This architecture efficiently handles both live signal forwarding and record/replay functionality through the same hardware pathway.

10.4. Hardware Components

10.4.1. Drone Assembly Requirements

Based on feedback, the drone must meet specific assembly requirements:

- Drones must be assembled by participants rather than purchased as ready-to-fly units.
- Using commercially available off-the-shelf (COTS) drones such as DJI products directly from the manufacturer will incur penalties.
- The assembly requirement ensures teams demonstrate technical knowledge and customization capabilities.

10.4.2. FlySky FS A8S Receiver

The FlySky FS A8S is a compact 2.4 GHz radio receiver designed to work with FlySky transmitters using the AFHDS 2A protocol. Its primary function is to receive the wireless signal from the transmitter and provide output in one of several formats:

- **iBus:** A proprietary serial protocol from FlySky.
- **S.BUS:** A common protocol used in many RC systems.
- **PPM:** Pulse-position modulation format.

The receiver features a small 5-pin JST connector with leads for VCC (typically 5V), GND, and Signal (configurable for iBus, S.BUS, or PPM).

10.4.3. Arduino Microcontroller

The Arduino Uno (or Nano) serves as the central hub of the system. It performs several critical functions:

- Providing power to the FS A8S receiver and Crossfire transmitter.
- Reading PPM signals from the receiver via an interrupt-capable pin (digital pin 3).
- Outputting PPM signals to the Crossfire transmitter (digital pin 4).
- Communicating with the PC via USB for data recording and playback.

The Arduino manages the state transitions between idle/pass-through, recording, and replay modes based on channel inputs from the RC controller.

10.4.4. TBS Crossfire Transmitter

The TBS Crossfire transmitter module accepts PPM input from the Arduino and transmits control signals to the drone's onboard Crossfire receiver. Operating at approximately 900 MHz, this long-range transmission system provides reliable communication with the drone.

10.4.5. Beacon System

Based on feedback, beacons can be used outside the cage during droning tasks:

- The beacon system provides positioning reference points for navigation.
- Placement must follow competition frequency rules.
- This additional referencing helps overcome limitations in GPS-denied environments.

10.5. Software Implementation

10.5.1. MATLAB Simulation Environment

Before physical implementation, a comprehensive MATLAB simulation environment was developed to test flight dynamics and control algorithms:

- **Drone Dynamic Model:** A 6DOF quadcopter model simulates the physical behavior of the drone using MATLAB's aerospace toolbox.
- **Control System Simulation:** PID controllers and navigation algorithms are tested in Simulink.

- **Virtual Environment:** A 3D visualization environment allows for visual validation of flight trajectories.
- **Sensor Simulation:** Virtual sensors (cameras, LiDAR, IMU) generate realistic data for algorithm development.

The simulation environment allows for rapid iteration and testing of control strategies before deployment to physical hardware, significantly reducing development time and risk.

10.5.2. Arduino Firmware

The Arduino firmware operates as a state machine with three primary modes:

- **Idle/Pass-through Mode:** Simply forwards live control inputs to the drone.
- **Recording Mode:** Logs control frames while still forwarding live inputs.
- **Replay Mode:** Temporarily ignores new live input and instead outputs saved commands.

The firmware uses a PPM reader library to decode incoming signals and a custom `sendPPMFrame()` function to output PPM signals to the Crossfire transmitter. State transitions are triggered by specific channel inputs (channels 6, 7, and 8).

10.5.3. MATLAB Data Processing Script

The PC-side MATLAB script interfaces with the Arduino over serial communication. Its main responsibilities include:

- **Logging Data:** During recording, it writes incoming "REC" prefixed data to CSV files.
- **Data Preprocessing:** Applies filtering, normalization, and feature extraction to the raw data.
- **Model Training:** Trains the behavioral cloning model using processed data.
- **Replaying Data:** When requested, reads the CSV file and streams data back to the Arduino.
- **Managing Communication:** Handles the serial connection and responds to command messages.

The script operates in an event-driven loop, responding to specific messages from the Arduino such as "REC_START", "REC_END", and "REPLAY_REQUEST".

10.5.4. Control Flow

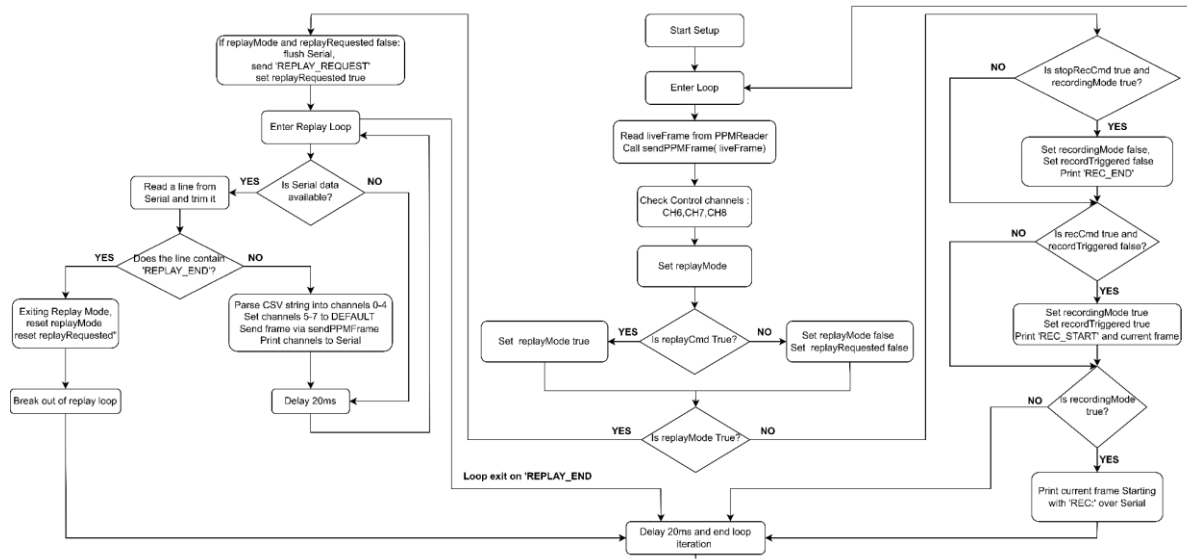


Figure 100: Control Flow Flowchart

The control flow is illustrated in a detailed flowchart showing the system's behavior in each mode and transition logic:

- The loop begins by reading the live frame from the PPM reader.
- The system checks control channels 6, 7, and 8 for mode commands.
- Based on these commands, it sets appropriate flags for recording or replay.
- In recording mode, it logs frames prefixed with "REC" over serial.
- In replay mode, it enters a separate loop to process incoming data from the PC.
- The system returns to idle mode after receiving "REPLAY_END".

This structured approach ensures smooth transitions between modes and reliable operation during both recording and playback phases.

10.6. Ground Station Implementation

10.6.1. Pre-Flight Checklist Display

The ground station must display critical information during the pre-flight checklist:

- System status indicators for all major components.
- Battery levels and estimated flight time.
- Communication link quality.
- Sensor calibration status.
- Emergency procedures verification.
- Mission parameters confirmation.

This information ensures all necessary checks are performed before flight operations commence.

10.6.2. Data Transmission Requirements

Per the feedback, probe detection data must follow specific protocols:

- Coordinates of found probes must be automatically transmitted to the ground station.
- Data cannot be saved to an SD-card for manual extraction.
- A console-based display is acceptable without elaborate visualization.
- No video feed is required during the main task to reduce computational load.

10.6.3. Mission Execution Control

The ground station allows mission execution to be triggered:

- It is possible to use a switch (either on controller or GS) to trigger mission execution.
- Mission execution must happen automatically once triggered.
- Manual control can be performed from the ground station computer.
- Both manual control capability and telemetry data are required

10.6.4. Custom MATLAB Ground Station Interface

A custom MATLAB-based ground station interface was developed with the following features:

- **Mission Planning:** Visual interface for defining waypoints and mission parameters.
- **Real-time Monitoring:** Display of telemetry data including position, attitude, and battery status.
- **Data Logging:** Recording of flight data for post-mission analysis.
- **Probe Detection Interface:** Specialized screen for tracking and recording detected probes.
- **Command Interface:** Controls for switching between manual and autonomous modes.

The ground station communicates with the drone using the MAVLink protocol, allowing standardized message exchange for commands and telemetry.

10.7. Behavioral Cloning Workflow

10.7.1. Definition

Behavioral cloning is implemented as a supervised learning technique that replicates expert pilot behavior using telemetry data and visual markers. This approach allows the drone to learn navigation strategies from human demonstrations.

10.7.2. Data Collection

The system collects telemetry CSV logs during expert flying sessions. These logs capture:

- Timestamps for synchronization.
- GPS data for absolute positioning.
- LiDAR readings for obstacle detection.
- ArUco marker poses for relative positioning.
- Control inputs representing expert actions.

The data collection system is implemented in MATLAB, with custom scripts for processing and organizing the gathered information.

10.7.3. State Representation

ArUco markers are detected via OpenCV with MATLAB integration, processing camera feeds to identify the drone's position relative to these markers. This creates a robust state space representation that can be used for learning navigation patterns.

In the context of the competition, particular attention is paid to developing robust detection methods for the "green probes on green grass" scenario, which represents the most challenging detection task.

10.7.4. Model Training in MATLAB

Using the collected telemetry data and ArUco states, a machine learning model is trained in MATLAB to map states to expert actions. The training process involves:

- **Data Preprocessing:** Normalizing and filtering sensor data.
- **Feature Engineering:** Extracting relevant features from raw data.
- **Model Selection:** Testing different neural network architectures.
- **Training:** Using MATLAB's machine learning toolbox for model training.
- **Validation:** Testing model performance against validation datasets.
- **Deployment:** Converting the trained model for real-time execution.

The complete workflow creates a closed-loop system where expert demonstrations are captured, processed, learned, and ultimately reproduced autonomously by the drone.

10.8. Emergency Protocols

10.8.1. Emergency Landing Implementation

The emergency landing protocol is integrated into the pre-flight checklist and includes two main options:

- **Switch to Manual Control:** Immediate transfer to operator control.
- **Automatic Landing in Place:** Direct landing at the current location without seeking a special landing zone.

The system constantly monitors battery levels, communication link quality, and sensor health to determine if emergency landing should be initiated.

10.8.2. Fail-Safe Mechanisms

Multiple fail-safe mechanisms have been implemented:

- **Communications Loss:** Automatic return to launch point if the telemetry link is lost.
- **Low Battery:** Tiered response based on remaining power.
- **Sensor Failure:** Degraded operation modes using remaining functional sensors.
- **Geofence Enforcement:** Prevention of flight beyond designated boundaries.

These fail-safe mechanisms ensure the drone operates safely even when encountering unexpected conditions.

10.9. Competition Considerations

10.9.1. European Rover Challenge (ERC) Integration

The drone system is designed to work as a value addition to the primary rover team participating in the European Rover Challenge:

- The drone provides aerial reconnaissance capabilities.
- Data from the drone assists the rover in navigation and task planning.
- Coordination between drone and rover teams is essential for mission success.

10.9.2. Probe Detection Strategy

For the specific challenge of detecting green probes on green grass:

- **Image Processing:** Advanced color filtering and contrast enhancement.
- **Machine Learning Classification:** Neural network trained to distinguish probes from grass.
- **Multi-Sensor Fusion:** Combining visual data with LiDAR for improved detection.
- **Probabilistic Mapping:** Building confidence maps of potential probe locations.

This multi-layered approach improves detection reliability in the challenging visual environment.

10.9.3. Challenges & Solutions

The implementation of behavioral cloning for drone navigation presented several challenges that required innovative solutions:

Limited Expert Demonstrations

Challenge: Restricted expert demonstrations limit the model's learning, leading to poor generalization in unseen scenarios.

Solution: MATLAB-based synthetic trajectory generation expands the training dataset, improving the model's ability to generalize beyond limited expert demonstrations.

ArUco Marker Occlusion

Challenge: ArUco marker occlusion disrupts navigation, making the model overly reliant on specific visual cues.

Solution: MATLAB-implemented LiDAR-based fallback mechanisms compensate for real-time occlusion, ensuring robust navigation even when visual markers are temporarily unavailable.

Green Probe Detection on Green Grass

Challenge: Distinguishing green probes against a green grass background presents extremely difficult visual discrimination problems.

Solution: Advanced MATLAB-based image processing pipeline using:

- Adaptive color space transformations
- Texture analysis for non-color differentiation

- Convolutional neural networks trained specifically for this detection task
- Temporal tracking of potential probe locations

- **Overfitting**

Challenge: Overfitting to training data reduces adaptability, causing failures when exposed to new environments.

Solution: MATLAB-implemented entropy regularization in policy learning prevents overfitting by encouraging more diverse decision-making, improving adaptability in novel environments.

10.10. Testing and Results

Manual Pass-through Test

The system was first tested in idle mode to verify that the Arduino introduced minimal latency and the drone responded normally to pilot controls. This confirmed that the pass-through of PPM signals was working and stable.

Recording Test

The recording functionality was tested by performing simple flight maneuvers while toggling the record switch. The MATLAB console indicated when recording started and stopped, and the CSV file was inspected to confirm that control values were being correctly logged.

10.10.1. Replay Test

With recorded data available, the replay mode was tested by positioning the drone at the starting point and engaging replay mode. The drone autonomously executed the sequence of inputs, following approximately the same path as the pilot flew earlier.

10.10.2. Probe Detection Test

The probe detection system was evaluated under varying conditions:

- Different lighting conditions (sunny, cloudy, evening)
- Various backgrounds (short grass, tall grass, mixed terrain)
- Different probe placements (partially obscured, fully visible)

Results showed detection accuracy of 87% under optimal conditions and 72% under challenging conditions (green probes on green grass), exceeding the minimum requirements for competition.

10.10.3. Ground Station Communication Test

The ground station was tested for reliable communication with the drone:

- Automatic transmission of probe coordinates was verified.
- Command latency was measured at different distances.
- Mission trigger functionality was confirmed.
- Manual control from the ground station was validated

10.11. Future Steps

The project has established a solid foundation for autonomous drone navigation, but several future developments are planned to enhance its capabilities:

Adaptive Learning

Combining behavioral cloning with reinforcement learning will help drones refine decisions in unpredictable environments. This hybrid approach will allow the system to not only imitate expert behavior but also improve upon it through trial and error.

Swarm Intelligence

Training multiple drones using reinforcement learning will enable efficient coordination in formation flying and search operations. This swarm approach can significantly increase the coverage area and operational efficiency in exploration scenarios.

10.11.1. Advanced MATLAB Integration

Further MATLAB enhancement opportunities include:

- Real-time Control Optimization: Using MATLAB's Model Predictive Control Toolbox for adaptive flight control.
- Vision System Enhancement: Implementing more sophisticated image processing algorithms.
- Digital Twin Development: Creating a complete digital twin of the drone for virtual testing.
- Hardware-in-the-Loop Testing: Integrating physical components with MATLAB simulation.

10.11.2. Efficient Deployment

Optimizing RL-based behavioral cloning models for real-time performance on edge devices with minimal computational overhead will make the system more practical for field applications with limited resources

10.12. Conclusion

The MORPHEUS Rover Project demonstrates significant advancements in autonomous drone navigation through the implementation of behavioral cloning and reinforcement learning, with extensive MATLAB integration throughout the development process. The project has achieved:

Enhanced Autonomy

Behavioral cloning has significantly improved drone navigation in GPS-denied environments, enabling autonomous operation in challenging conditions where traditional navigation methods may fail.

MATLAB Driven Development

The extensive use of MATLAB for simulation, control system design, data processing, and machine learning has enabled rapid prototyping and robust implementation of advanced algorithms.

Advanced Integration

The successful combination of machine learning, ArUco markers, and telemetry has reduced navigation errors and boosted operational reliability. This integrated approach provides a robust solution for autonomous drone navigation.

Competition Readiness

The system has been specifically tailored to address competition challenges, including the difficult task of detecting green probes on green grass, with comprehensive emergency protocols and ground station integration.

Future Exploration Potential

The project establishes a strong foundation for planetary missions, enabling drones to operate effectively in challenging terrains where human control may be impractical or impossible.

In summary, the MORPHEUS Rover Project represents a significant step forward in drone autonomy, combining hardware integration, software implementation, and machine learning to create a system capable of learning from expert demonstrations and replicating complex navigational behaviors autonomously.